A P A L I N

DESCRIPTION:
APALIN 100 MG INJECTION (2 ml amp): A clear, almost colourless liquid.
APALIN 250 MG INJECTION (2 ml amp): A clear, very pale straw coloured liquid.
APALIN 500 MG INJECTION (2 ml amp): A clear, pale straw coloured liquid.

COMPOSITION:
APALIN 100 MG INJECTION (2 ml amp): Each ml contains Amikacin Sulfate equivalent to Amikacin 50 mg. Each ampoule contains Amikacin Sulfate equivalent to 100 mg Amikacin per 2 ml.
APALIN 250 MG INJECTION (2 ml amp): Each ml contains Amikacin Sulfate equivalent to Amikacin 125 mg. Each ampoule contains Amikacin Sulfate equivalent to 250 mg Amikacin per 2 ml.
APALIN 500 MG INJECTION (2 ml amp): Each ml contains Amikacin Sulfate equivalent to Amikacin 250 mg. Each ampoule contains Amikacin Sulfate equivalent to 500 mg Amikacin per 2 ml.

PHARMACODYNAMICS:
Class: Semi-synthetic aminoglycoside; broad spectrum antibiotic.
Relationship to other drugs: Amikacin is an analogue of kanamycin. The molecule has been structurally altered to render it resistant to 8 of the 9 possible routes of enzymatic deactivation by bacteria.
Site and Mode of Action: It exerts its bactericidal action by inhibiting protein synthesis in bacterial ribosomes and hence interferes with the reproduction of the genetic code.

PHARMACOKINETICS:
Absorption: After IM injection amikacin is rapidly absorbed. In normal adult volunteers, average peak serum concentrations of 12, 16 and 21 microgram/mL are obtained 1 hour after IM injection of 250 mg (3.7 mg/kg), 375 mg (5 mg/kg) and 500 mg (7.5 mg/kg) single doses respectively. After 10 hours serum levels are about 0.3 micrograms/mL, 1.2 micrograms/mL and 2.1 micrograms/mL respectively. After IV administration of single doses of 500 mg (7.5 mg/kg to normal adults as an infusion over 30 minutes, a mean peak serum concentration of 38 micrograms/mL was reached at the end of the infusion period. Levels of 24 micrograms/mL, 18 micrograms/mL and 0.75 micrograms/mL were present after 30 minutes, 1 hour and 10 hours post-infusion respectively.
Bioavailability: Mean urinary recovery after IV or IM administration is 95-98% of the dose.
Distribution: The volume of distribution is about 24 litres. Amikacin penetrates bone and has been found in saliva, pleural fluid, synovial fluid and peritoneal fluid. It crosses non-inflamed meninges giving low CSF concentrations. It is excreted in the urine attaining high concentrations *(see **Excretion** below)*. Amikacin does not penetrate human red blood cells. It is not known whether it enters breast milk. In the foetus, peak serum concentration is about 16% of the peak maternal serum concentration. In neonates, it is thought that the drug remains primarily in the extracellular fluid. Besides in newborns of varying weights at a dose of 7.5 mg/kg showed that serum half-life values are correlated inversely, with post-natal age and renal clearance of amikacin. Spinal fluid levels in normal infants are approximately 10 to 20% of serum concentrations and may reach 50% when the meninges are inflamed.
Protein binding: The mean human serum protein binding (determined by ultra-centrifugation) of amikacin is 11.2% ± 8% over a concentration of 5 – 50 micrograms/mL.
Metabolism: Amikacin is excreted unchanged.
Excretion: Amikacin is excreted in the urine primarily by glomerular filtration. With normal renal function about 90% of an IM dose is excreted unchanged in the urine in the first 8 hours and 98% within 24 hours. Mean urine concentrations for the first 6 hours are 563 micrograms/mL following a 250 mg dose, 697 micrograms/mL following a 375 mg dose and 932 micrograms/mL following a 500 mg dose. The renal clearance rate is 95 mL/min in normal renal function. Patients with impaired renal function excrete the drug more slowly.
Half-life: Amikacin has a mean half-life of about 2 hours in patients with normal renal function. This is prolonged in patients with impaired renal function. The serum half-life is longer in neonates *(see **Paediatric Dosage**)*.
Clinical Significance of Pharmacokinetics Data: Amikacin in a dose of 15 mg/kg/day divided into 2 or 3 equal doses given at equi-spaced time intervals provides adequate blood levels. As it is excreted unchanged by glomerular filtration, impairment of renal function may result in accumulation of the drug and the possibility of ototoxicity, nephrotoxicity and neurotoxicity becomes greater. Therefore the dose must be reduced in these patients *(see **Dosage in Impaired Renal Function**)*.

INDICATIONS:
1. Short term treatment of serious infections due to susceptible strains of gram negative bacteria (see below for susceptible strains).
2. Treatment of staphylococcal infections: The use of amikacin in the treatment of staphylococcal infection should be restricted to second line of defence rather than initial therapy and should be confined to patients suffering from severe infection caused by susceptible strains of staphylococcus who have failed to respond or are allergic to other available antibiotics.
3. Treatment of neonatal sepsis when sensitivity testing indicates that other aminoglycosides cannot be used. In certain severe infections such as neonatal sepsis, concomitant therapy with a penicillin-type drug may be indicated because of the possibility of infections due to gram-positive organisms such as streptococci or pneumococci.

Clinical studies has shown amikacin to be effective in the following infections when due to susceptible organisms; bacteraemia, septicaemia, neonatal sepsis; serious infections of the respiratory tract, bones and joints, central nervous system, skin and soft tissues, intra-abdominal organs; burns, postoperative infections and recurrent urinary tract infections.

Antibacterial activity: Amikacin is active, in vitro, against the following organisms.

	M.I.C. Micrograms/ mL
Gram-negative	
Pseudomonas species	1.6 - 12.5
Escherichia coli	1.6 - 3.1
Proteus species (indole-positive and indole negative)	1.6 - 3.1
Providencia stuartii	3.1
Klebsiella pneumoniae	1.6 - 3.1
Enterobacter cloacae	1.6 - 3.1
Serratia species	0.8 - 3.1
Citrobacter freundii	1 - 8
Acinetobacter	No information available
Gram-positive	
Staphylococcus species (penicillinase and non-penicillinase producing, including methicillin resistant strains)	0.4 - 5.0

RECOMMENDED DOSAGE:
The status of renal function should be estimated by serum measurement of the serum creatinine concentration or calculation of the endogenous creatinine clearance rate. The blood urea nitrogen (BUN) is much less reliable for this purpose. Reassessment of renal function should be made periodically during therapy. Whenever possible, amikacin concentrations in serum should be measured to assure adequate but not excessive levels. It is desirable to measure both peak and trough serum concentrations intermittently during therapy. Peak concentrations (30-90 minutes after injection) above 35 mcg per mL and trough concentrations (just prior to the next dose) above 10 mcg per mL should be avoided. Dosage should be adjusted as indicated.
Adult: Dosages are calculated on a body weight basis. The following dosages are for patients with normal renal function.
IM Administration: Recommended total daily dose = 15 mg/kg/day divided into 2 or 3 equal doses administered at equispaced intervals: eg. 7.5 mg/kg every twelve hours or 5 mg/kg every eight hours. The maximum recommended daily dose is 15 mg/kg/day and in heavier patients, 1.5 g/day. When amikacin sulphate is indicated in uncomplicated urinary tract infections, a dose of 250 mg twice daily may be used. Usual duration of treatment = 7-10 days. If treatment for longer than 10 days is considered, the use of amikacin should be re-evaluated and if continued, renal and auditory functions should be monitored daily. At the recommended dosage level, uncomplicated infections due to amikacin sensitive organisms should respond in 24 to 48 hours. If definite clinical response does not occur within 3 to 5 days, therapy should be stopped and the antibiotic sensitivity pattern of the invading organisms should be rechecked. Failure of the infection to respond may be due to resistance of the organism or to the presence of septic foci requiring surgical drainage.
IV administration: Recommended total daily dose is 15 mg/kg/day. The maximum daily dose and duration of treatment are identical to that recommended for IM administration. The IV infusion is prepared by adding the 500 mg/2 mL amikacin solution to 200 mL of either 0.9% Sodium Chloride or 5% Glucose in water. Administer over 30-60 minute period. The total daily dose should not exceed 15 mg/kg/day and should be given in 2 or 3 equally divided doses at equispaced time intervals. Do not physically premix amikacin injection or infusion solutions with other drugs at any point in the infusion apparatus. It should be administered separately according to the recommended dose and route.

Paediatric: The dosages given are for patients with normal renal function.

New born infants: IM Administration: Initiate treatment with an initial dose of 10 mg/kg and continue therapy with 7.5 mg/kg every 12 hours. Amikacin has a half-life of 7 to 9 hours at birth due to immaturity of renal excretion mechanisms. Half-life decreases with increasing age to reach 3 hours at 1 month of age. Care should be taken to calculate the doses accurately and the 50 mg/mL reconstituted solution should be further diluted when necessary to allow administration of accurate doses in the smaller premature. Whenever possible serum levels of amikacin should be monitored to allow the dose adjustments needed to maintain amikacin concentrations within the therapeutic range.

IV administration: Use the same dose as given under IM administration. The infusion should be given over 1-2 hours. Reconstitute injection as recommended under adult IV administration.

Children and older infants: IM Administration: 15 mg/kg/day divided into 2 or 3 equal doses and administered at equispaced time intervals. Maximum total daily dose should not exceed 15 mg/kg/ day. The recommendations concerning duration of therapy are the same as for adult dosages.

IV administration: Use dosages given under IM administration. The amount of fluid used for the infusions will depend on the amount ordered for the patient. It should be a sufficient amount to infuse the amikacin over a 30- 60 minute period.

Geriatric: Use adult dose with caution. See dosage below in Impaired Renal Function if applicable.

In Impaired Liver Function: As amikacin is excreted unchanged in the urine, impairment of liver function should not require any change in dosage of the drug.

In Impaired Renal Function: Dosage of amikacin needs to be reduced in patients with impairment. Whenever possible, serum amikacin concentrations should be monitored by appropriate assay procedures. Doses may be adjusted either by administering normal doses at prolonged intervals or by administering reduced doses at a fixed interval. Both methods should be based on the patient's creatinine clearance or serum creatinine values since these have been found to correlate with aminoglycoside half-lives in patients with diminished renal function. These dosage schedules must be used in conjunction with careful clinical and laboratory observations of the patient and should be modified as necessary. Neither method should be used when dialysis is being performed.

Normal Dosage at Prolonged Intervals: If the creatinine clearance is not available and the patient's condition is stable, a dosage interval in hours for the normal dose can be calculated by multiplying the patient's serum creatinine by nine, e.g. if the serum creatinine concentrations is 2 mg/100 mL the recommended single dose (7.5 mg/kg) should be administered every 18 hours.

Reduced Dosage at Fixed Time Intervals: If it is desired to administer Amikacin at a fixed time interval, the dosage must be reduced. It is recommended that the drug be given every 12 hours. In these patients, serum concentrations of the drug should be measured to assure accurate administration and avoid concentrations of amikacin above 35 micrograms/mL. If serum assay concentrations are not available and the patient's condition is stable, serum creatinine and creatinine clearance values are the most readily available indicators of the degree of renal impairment to use as a guide for dosage. However, in elderly patients creatinine clearance values should be used in preference to serum creatinine values so that a more accurate estimation of renal function may be made. First, initiate therapy by administering half the normal daily dose (7.5 mg/kg) which would be calculated for a patient with a normal renal function as described above. To determine the size of maintenance doses administered every 12 hours, the initial dose should be reduced in proportion to the reduction in the patient's creatinine clearance rate:

$$\text{Maintenance Dose Every 12 hours} = \frac{\text{Observed cc in Ml / min}}{\text{normal cc in mL/min}} \times \text{Calculated initial dose in mg}$$

An alternative rough guide for determining reduced dosage at twelve-hour intervals (for patients whose steady state serum creatinine values are known) is to divide the normally recommended dose by the patient's serum creatinine. The above dosage schedules are not intended to be rigid recommendations, but are provided as guides to dosage when the measurement of amikacin serum levels is not feasible.

Stability: Apalin 250mg/2ml and 500mg/2ml Injection remained stable for up to 24 hours at 30°C, and up to 60 days at 2-8°C, after diluted with Sodium Chloride 0.9% Solution (Normal Saline), Dextrose 5% Solution, or Lactated Ringer's Solution inside a Polypropylene Syringe. Solutions may vary in color from colorless to light straw or very pale yellow; this variation does not affect their potency. Discard dark-colored solutions.

Preparation of dosage form: To prepare initial dilution for intravenous use, add the contents of each 500-mg vial to 100 to 200 mL or each 250-mg vial to 50 to 100mL of 0.9% sodium chloride injection, 5% dextrose injection, Lactated Ringer's Solution, or other suitable diluent. The resulting solution should be administered slowly over a 30- to 60-minute period to help avoid neuromuscular blockade. Pediatric patients may require a proportionately smaller volume of diluent.

ROUTE OF ADMINISTRATION : For I.V and I.M administration

CONTRAINDICATIONS:

- History of hypersensitivity to amikacin
- Pregnancy
- Lactation
- History of hypersensitivity or serious toxic reactions to aminoglycosides

WARNINGS AND PRECAUTIONS:

Safety for treatment periods longer than 14 days has not been established. Ototoxicity, both auditory and vestibular can occur in patients treated at higher doses or for periods longer than those recommended. The risk of amikacin-induced ototoxicity is greater in patients with renal damage. High frequency deafness usually occurs first, and can be detected only by audiometric testing. Vertigo may occur and may be evidence of vestibular injury. Patients receiving amikacin should therefore be under close clinical observation for signs of toxicity which if it occurs requires discontinuation of the drug. Lost function may not be fully restored.

Nephrotoxicity: Aminoglycosides are potentially nephrotoxic. Renal function should be closely monitored in patients with known or suspected renal impairment, and also in those whose renal function is initially normal but who develop signs of renal dysfunction during therapy. Since amikacin is present in high concentrations in the renal excretory system, patients should be well hydrated to minimize chemical irritation of the renal tubules. Kidney function should be assessed by the usual methods prior to starting therapy and daily during the course of treatment. If signs of renal irritation appear (casts, white or red cells or albumin) hydration should be increased. A reduction in dosage (see Dosage in Impaired Renal Function) may be desirable if other evidence of renal dysfunction occurs such as decreased creatinine clearance, decreased urine specific gravity, increased BUN, increased creatinine, or oliguria. If azotaemia increases or if a progressive decrease in urinary output occurs, treatment should be stopped.

Neurotoxicity: Neurotoxicity, manifested as vestibular and permanent bilateral auditory ototoxicity, can occur in patients with pre-existing renal damage and in patients with normal renal function, treated at higher doses and/or for periods longer than recommended. Other manifestations of neurotoxicity may include numbness, skin tingling, muscle twitching and convulsions. The risk of hearing loss due to aminoglycosides increases with the degree of exposure to either high-peak or high trough serum concentrations. Patients developing cochlear damage may not have symptoms during therapy to warn them of developing eighth nerve toxicity, and total or partial irreversible bilateral deafness may occur after the drug has been discontinued. Aminoglycoside-induced ototoxicity is usually irreversible. Neuromuscular blockade and muscular paralysis have been demonstrated in the cat with high doses of amikacin (188 mg/kg). The possibility of neuromuscular blockade and respiratory paralysis should be considered when amikacin is administered concomitantly with anaesthetic or neuromuscular blocking drugs. If blockade occurs, calcium salts may reverse this phenomenon, but mechanical respiratory assistance may be necessary.

Use in newborn infants: The ototoxic and nephrotoxic potential of amikacin in infants is not known. Until more safety reports become available, amikacin should be used in infants only in those specific circumstances when susceptibility testing indicates that other aminoglycosides cannot be used or are otherwise contra-indicated, and when the infant can be observed closely for evidence of toxicity.

Sulphite sensitivity: Amikacin contains sodium bisulphite, a sulphite that may cause allergic-type reactions including anaphylactic symptoms and life-threatening or less severe asthmatic episodes in certain susceptible people. The overall prevalence of sulphite sensitivity in the general population is not known but is probably low. Sulphite sensitivity is seen more frequently in asthmatic than in nonasthmatic subjects.

Precautions: Cross allergenicity has been demonstrated among aminoglycoside antibiotics. Therefore caution should be exercised in patients who have shown hypersensitivity to this class of drugs.

Superinfection: As with other antibiotics, the use of amikacin may result in overgrowth of nonsusceptible organisms. If this occurs appropriate therapy should be instituted.

Monitoring of serum levels: To lessen the risk of ototoxicity, nephrotoxicity and neurotoxicity, serum concentrations should be monitored when feasible, and prolonged peak concentrations above 35 micrograms/mL should be avoided.

Anaesthesia: The possibility of neuromuscular blockade and respiratory paralysis should be borne in mind when administering anaesthetics or neuromuscular blocking drugs.

INTERACTIONS WITH OTHERS MEDICAMENTS:

Concurrent Use with Other Antibiotics: Concurrent and/or sequential use of topically or systemically neurotoxic or nephrotoxic antibiotics, particularly bacitracin, cisplatin, amphotericin B, kanamycin, gentamycin, tobramycin, neomycin, streptomycin, cephaloridine, paromomycin, viomycin, polymyxin B, colistin, and vancomycin should be avoided.
Concurrent Use with Potent Diuretics: Amikacin should not be given concurrently with potent diuretics (ethacrynic acid, frusemide, meralluride sodium, sodium mercaptomerin, or mannitol). Some diuretics themselves cause ototoxicity, and intravenously administered diuretics enhance aminoglycosides toxicity by altering antibiotic concentrations in serum and tissue.

USE IN PREGNANCY AND LACTATION:

Use in Pregnancy:
Gentamicin and other aminoglycosides are known to cross the placenta. There is evidence of selective uptake of gentamicin by fetal kidney resulting in cellular damage (probably reversible) to immature nephrons. Eighth cranial nerve damage has also been reported following in-utero exposure to some of the aminoglycosides. Because of their chemical similarity, all aminoglycosides must be considered potentially nephrotoxic and ototoxic to the fetus. It should also be noted that therapeutic blood levels in the mother do not equate with safety for the fetus. There are no well-controlled studies in pregnant women, but investigational experience does not include any positive evidence of adverse effects to the foetus. Although there is no clearly defined risk, such experience cannot exclude the possibility of infrequent or subtle damage on the foetus.
Reproduction studies have been performed in rats and mice, and have revealed no evidence of impaired infertility or harm to the foetus due to amikacin. Amikacin should only be used in pregnant women when the expected benefit clearly outweighs the potential risk.
Use in Lactation:
It is not known whether this drug is excreted in breast milk. Since the possible harmful effect on the infant is not known it is recommended that if amikacin must be given to nursing mothers, the infant should not be breast fed during therapy.

SIDE EFFECTS:

The percentages below refer to incidence in clinical trials.
More common reactions:
Auditory and vestibular: hearing loss (4%) (permanent in some cases)
Biochemical Abnormalities: Increased BUN, elevated serum creatinine, decreased creatinine clearance, azotaemia.
Genito-urinary: reduced renal function.
Injection site reactions: pain at site of IM injection (6%)
Less common reactions:
Auditory and vestibular: tinnitus, vertigo, dizziness, nystagmus, changes in caloric testing or electrostagmograms.
Biochemical Abnormalities: Casts in urine, eosinophilia, increase in SGOT.
Dermatological: rash, pruritus. **Gastrointestinal**: nausea, vomiting **General**: drug fever
Genito-urinary: renal failure **Haematological**: anaemia. **Musculo-skeletal**: arthralgia
Nervous System: paraesthesia, headache, tremor
Serious or life-threatening reactions: Ototoxicity and renal toxicity are dose-related and are more likely to occur if the total cumulative dose exceeds 15 g.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Clinical features: Likely signs and symptoms include tinnitus, vertigo, reversible or irreversible deafness, skin rash, drug fever, headache, paraesthesia, reduced renal function or renal failure.
Management: Removal of the drug by peritoneal dialysis or haemodialysis will prevent further damage. In some cases, renal failure may need to be treated. In the newborn infant, exchange transfusion may also be considered.

INCOMPATIBILITIES:

Amikacin is incompatible with some penicillins and cephalosporins, amphotericin chlorothiazide sodium, erythromycin gluceptate, heparin, nitrofurantoin sodium, phenytoin sodium, thiopentone sodium and warfarin sodium, and depending on the composition and strength of the vehicle, tetracyclines, vitamins of the B group with vitamin C, and potassium chloride.
At times, amikacin may be indicated as concurrent therapy with other antibacterial agents in mixed or superinfections. In such instances, amikacin should not be physically mixed with other antibacterial agents in syringes, infusion bottles or any other equipment. Each agent should be administered separately.

STORAGE CONDITIONS:

Store below 30˚C. Protect from Light. Keep out of reach of children. *Jauhkan daripada kanak-kanak*
Note: At times the solution may darken from a colorless to be a pale yellow, but this does not indicate a loss in potency.

PACK SIZES:

Pack in 10’s ampoules of 2 ml / box.

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59, Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA